



SIMATS ENGINEERING



Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -1

C05_AT1

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1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

ANSWER :-

1. Recommended Internet Connectivity

- Primary Link: Low Earth Orbit (LEO) Satellite (e.g., Starlink Business / Eutelsat OneWeb).
- Secondary Link (Failover / Load Balancing): Dual-SIM 4G/5G Cellular Gateway with high-gain external directional antennas pointing toward the nearest telecom tower.

Justification:

- Bandwidth & Latency: LEO satellite delivers 100–220+ Mbps download speeds with low latency (25–45 ms), ensuring smooth video conferencing and responsive cloud application performance compared to high-latency legacy GEO satellites (600+ ms).
- Reliability: Two completely independent transmission mediums (satellite + terrestrial cellular) prevent downtime caused by weather events or single-carrier tower outages.
- Cost: Avoids the prohibitive capital expense of laying private fiber or dedicated microwave links to a rural site while maintaining standard monthly operational costs.
- Scalability: Additional satellite terminals or carrier SIMs can be added without infrastructural overhaul.

2. Networking Devices

- Edge Gateway / Router: SD-WAN Appliance with Integrated Next-Gen Firewall (NGFW) (e.g., Fortinet FortiGate 40F/60F or Cisco Meraki MX67/MX68).
- Local Area Network (LAN): Managed PoE+ Gigabit Switch (e.g., 24-Port Managed Switch) to power access points and VoIP devices with dedicated VLAN segmentation.

- Wireless LAN (WLAN): Wi-Fi 6 Enterprise Access Points (APs) (e.g., Aruba Instant On / Ubiquiti UniFi) configured with dedicated SSIDs (Corporate, Guest, IoT).

Justification:

- SD-WAN & Traffic Steering: SD-WAN dynamically routes latency-sensitive traffic (VoIP, Zoom/Teams) over the best-performing link, offloads general web traffic to cellular, and handles sub-second seamless failover.
- Cost Efficiency: Combining the router, SD-WAN controller, and firewall into a single appliance minimizes upfront hardware costs, power draw, and licensing fees.

3. Security Mechanisms

- Site-to-Site & Remote Access: IPsec VPN with IKEv2 / AES-256 encryption for an always-on tunnel back to HQ data centers, combined with a Zero Trust Network Access (ZTNA) client for direct cloud access.
- Edge Defense: NGFW Services enabling Intrusion Prevention System (IPS), SSL/TLS Deep Packet Inspection, DNS Filtering, and Antivirus at the network edge.

Summary Architecture Comparison

Component	Recommendation	Primary Justification
WAN 1	LEO Satellite	Low latency (25–45 ms), high bandwidth (100–220 Mbps), fast deployment
WAN 2	4G/5G Cellular Dual-SIM	Inexpensive, carrier-redundant backup path for uninterrupted uptime
Edge Router	SD-WAN + NGFW Appliance	Dynamic path selection, bandwidth optimization, sub-second failover
Security	IPsec VPN + ZTNA + 802.1X	End-to-end encryption to HQ, granular cloud access, and local segmentation

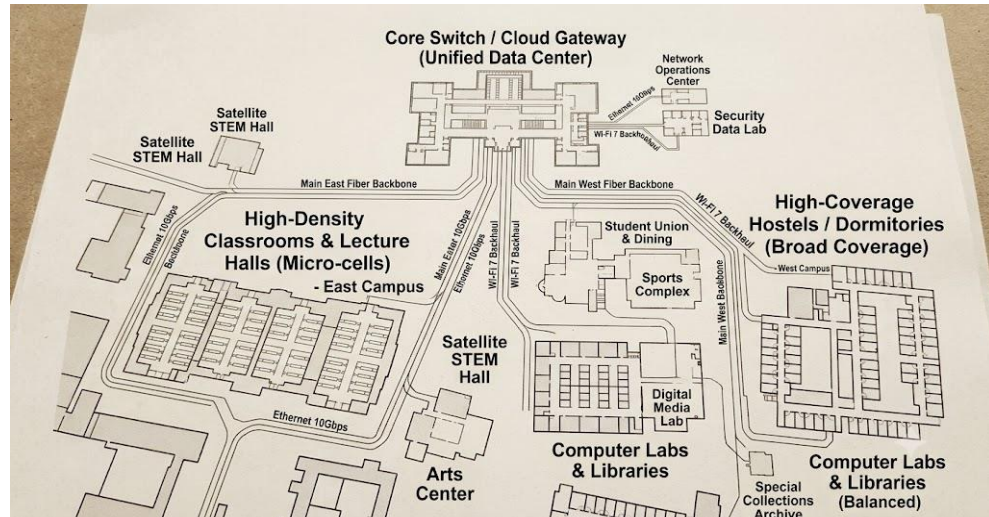
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

ANSWER :-

To provide reliable, secure campus-wide Wi-Fi for 3,000+ concurrent users with a limited number of Access Points (APs), the network must prioritize capacity-driven cell design, dual-band channel engineering, and centralized controller optimization.

1. Strategic AP Placement & Density Planning

Deploying a limited AP pool requires allocating hardware based on client density and throughput demand rather than uniform physical area.



- **Classrooms & Lecture Halls (High Density, High Concurrency):**
 - Deploy high-capacity Wi-Fi 6 (802.11ax) 4x4 MU-MIMO APs centered above seating areas.
 - Reduce AP transmit power (10–14 dBm) to create smaller RF cells. This prevents co-channel interference (CCI) while enabling spatial reuse (BSS Coloring).

- Libraries & Labs (Medium Density, High Throughput):
 - Place APs near study zones and open aisles, away from thick concrete pillars and metal book stacks to minimize attenuation.
- Hostels & Dorms (Dispersed, Wall Obstacles):
 - Place APs staggered along central hallways or deploy cost-effective in-wall APs (wall-plate APs) in alternate rooms to penetrate drywall/brick without line-of-sight dead zones.

2. Frequency Band Selection & Radio Frequency (RF) Management

To prevent spectrum exhaustion across 3,000+ simultaneous devices:

- 5 GHz Band (Primary Workhorse):
 - Channel Width: Standardize on 20 MHz channels (or selective 40 MHz in low-density zones). Avoid 80/160 MHz channels because wide channels drastically reduce the number of non-overlapping channels available, causing severe co-channel interference.
 - DFS Channels: Enable Dynamic Frequency Selection (UNII-2 / UNII-2 Extended) to unlock up to 25+ non-overlapping channels.
 - Role: Carry ~80% of client traffic (laptops, phones, streaming, submissions).
- 2.4 GHz Band (Coverage & Legacy Support):
 - Use strictly non-overlapping channels 1, 6, and 11 with 20 MHz width.
 - Disable 2.4 GHz radios on 50% of APs (or turn power down to 6–9 dBm) to prevent extreme 2.4 GHz overlap and beacon flooding.
- Optimizations:
 - Band Steering: Automatically steer 5 GHz-capable devices away from 2.4 GHz.
 - Minimum Basic Rate Enforcement: Disable legacy data rates below 12 Mbps. This drops slow, distant clients faster and prevents slow transmissions from consuming airtime.

- Fast Roaming: Implement 802.11k/v/r to ensure smooth handoffs between APs as students walk between buildings.

3. Authentication & Network Segmentation

A scalable, tiered authentication architecture balances student ease of use with enterprise security:

- Students & Faculty (Primary SSID: Campus-Secure):
 - Mechanism: WPA3-Enterprise / 802.1X with RADIUS / EAP-TLS (or PEAP-MSCHAPv2) integrated into the university's central identity store (LDAP / Active Directory / Google Workspace).
 - Advantage: Eliminates shared pre-shared keys (PSKs). Individual credentials prevent credential leakage and provide individual user accountability.
- Guests & Visitors (Secondary SSID: Campus-Guest):
 - Mechanism: Captive portal with SMS/Email OTP verification, client isolation enabled, and automatic session timeouts (e.g., 2 hours).
- Campus IoT & Lab Equipment (Hidden SSID: Campus-IoT):
 - Mechanism: WPA3-Personal / MPSK (Multi-Pre-Shared Key) with device MAC-address filtering.

4. Architectural Balance: Coverage, Performance, Security & Cost

Dimension	Implementation Strategy	Impact / Trade-Off Managed
Performance	20 MHz 5 GHz channels + Wi-Fi 6 OFDMA & MU-MIMO + Disabling rates < 12 Mbps	Maximizes airtime efficiency; supports high concurrent user density per AP without channel contention.
Coverage	Hybrid placement (high-density microcells in auditoriums; staggered APs in dorms)	Eliminates blackouts in critical study zones while covering

Dimension	Implementation Strategy	Impact / Trade-Off Managed
		expansive hostel spaces economically.
Security	802.1X / WPA3-Enterprise + Dynamic VLAN assignment per user role	Strong per-user encryption, automated role-based access control, and complete isolation of guest traffic.
Cost	Centralized Cloud/Virtual Wireless Controller + selectively disabling unused radios	Eliminates expensive physical controller appliances; reduces hardware requirements through optimized RF reuse.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

ANSWER :-

1. Network Segmentation & Topology

A multi-tier Zero-Trust Network Segmentation model isolates customer databases from less secure segments while controlling lateral movement:

- Demilitarized Zone (DMZ): Public-facing web/mobile banking front-ends, reverse proxies, and external API gateways.

- **Application Tier:** Internal business logic and transaction processing engines; accessible strictly from the DMZ over hardened ports.
- **Core Database Tier (High-Security Enclave):** Core banking servers and PCI-DSS sensitive cardholder data environments (CDE). No direct internet access.
- **Corporate User Subnets:** Micro-segmented by role (Tellers, Loans, Back Office, IT/Admin) using 802.1Q VLANs and Access Control Lists (ACLs).
- **Out-of-Band Management (OOBM):** Dedicated, isolated VLAN accessible only via jump servers/bastion hosts with hardware tokens.

2. Firewall Deployment Strategy

- **Perimeter Firewall (HA Pair):** Next-Generation Firewall (NGFW) handling DDoS mitigation, TLS inspection, and strict ingress/egress filtering.
- **Internal Segmentation Firewalls (ISFW):** High-throughput, low-latency NGFW appliances placed between the Corporate LAN, DMZ, Application, and Database tiers.
- **Web Application Firewall (WAF):** Dedicated inspection at the DMZ layer targeting OWASP Top 10 vulnerabilities (SQLi, XSS, CSRF) on web banking portals.

3. Authentication & Access Control

- **Multi-Factor Authentication (MFA):** FIDO2/WebAuthn hardware security keys (e.g., YubiKey) for privileged admins; app-based push notifications with number matching for standard staff.
- **Role-Based & Context-Aware Access (RBAC / ABAC):** Least-privilege access controlled through centralized identity providers (e.g., Microsoft Entra ID / FreeIPA), restricting database access based on user role, device compliance state, and physical location.
- **Privileged Access Management (PAM):** Session recording, credential vaulting, and just-in-time (JIT) elevation for database and infrastructure administrators.

4. Intrusion Detection & Threat Monitoring

- **NIDS / NIPS:** Inline Next-Gen IPS on ISFWs for automated signature and anomaly blocking across inter-VLAN traffic, combined with passive tap-based NIDS (e.g., Zeek / Suricata) for deep traffic inspection without latency impact.

- Host-Level EDR / XDR: Deployed on all endpoints and servers to detect memory injection, unauthorized privilege escalation, and lateral movement.
- SIEM / SOC Integration: Centralized log aggregation (Elastic SIEM / Wazuh / Splunk) with real-time correlation against threat intelligence feeds to meet compliance audit requirements.

5. Architectural Balance Matrix

Requirement	Implementation	Justification & Trade-off Management
Security	Zero-Trust micro-segmentation, ISFWs, inline IPS, EDR	Prevents lateral spread if an employee workstation is compromised.
Performance	ASIC-accelerated firewalls, dedicated VLAN routing at core L3 switches	Offloads non-critical inter-VLAN routing to hardware; avoids inspection bottlenecks on routine traffic.
Compliance	Strict CDE isolation (PCI-DSS), encrypted audit logs (SOX/GLBA/GDPR)	Shrinks compliance audit scope by walling off core customer data into dedicated micro-segments.
Budget	Consolidated NGFW/ISFW licensing + open-source telemetry tools (Zeek/Wazuh)	Minimizes hardware footprint and vendor sprawl while avoiding expensive proprietary SIEM licensing.